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DISCUSSION PAPER

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Virtual age, is it real? - Discussing virtual age in reliability context

Maxim Finkelstein[1,2] | **Ji Hwan Cha**[3]

[1]Department of Mathematical Statistics, University of the Free State, Bloemfontein, South Africa

[2]ITMO University, St. Petersburg, Russia

[3]Department of Statistics, Ewha Womans University, Seoul, Republic of Korea

Correspondence
Ji Hwan Cha, Department of Statistics, Ewha Womans University, Seoul 120-750, Republic of Korea.
Email: jhcha@ewha.ac.kr

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Abstract

Several settings, where the notion of virtual age is employed, are discussed. We argue that the age reduction imperfect repair modeling is often not sufficiently justified in practice, as it is not possible to execute repair in most of real situations that conforms with this model. On the other hand, a shock-based virtual age model is suggested and justified via the probabilities of failures on shocks. The new notion of virtual age for degrading items is also introduced. We discuss how to recalculate virtual age after switching from one regime to another. Several examples illustrating different notions of virtual age are presented.

KEYWORDS

age reduction, degradation, imperfect repair, minimal repair, virtual age

1 | INTRODUCTION

We discuss an important notion of virtual age that is widely used in reliability literature, however, often without questioning the underlying assumptions and, therefore, practical applicability. Our main interest is in the corresponding fundamental, underlying properties in different applications, where this notion arises. We discuss *some* of the well-known results in the literature (including our previous publications on the subject) from a different viewpoint and *focus on possible justifications and comparisons* of the corresponding models and approaches, where possible (Sections 2, 3, and 6). We also suggest and analyze several new virtual age based settings (Sections 4, 5, and 7) that are aligned with our general approach and focus on some relevant misconceptions as well.

The purpose of this article is not in performing a survey on the topic (we refer only to those sources that are relevant to our reasoning), but rather to engage a discussion on this *somewhat controversial notion*. We understand that some of our arguments can be also considered as controversial, but this is one of the features of the discussion paper (as we see it) that invites to deliberation. In this brief introduction, we just outline a few key points to be developed in what follows.

There are two main "streams" in the literature, where the virtual age concept is often used in reliability context. The first one is when the repair action changes the state of a failed item, improving it to some intermediate level between "as good as new" (perfect repair) and "as bad as old" (minimal repair). In this way, perfect, minimal and imperfect repairs are interpreted in a natural way. The proxy for the "intermediate state" is some intermediate age, whereas perfect repair, obviously, sets this age to 0 and minimal repair does not change it. The age that describes the state of an item after the imperfect repair is often called in the literature the "virtual age." Kijima[1] and Finkelstein[2] were probably the first to use this term in the framework of maintenance actions. In order the virtual age, as the proxy of the intermediate state, to become a meaningful characteristic, certain rather stringent conditions should be imposed and we discuss this in detail in the next section. Our main argument is that these conditions are not usually met in practice and the real maintenance actions usually do not conform with them. Only minimal repair, which can be also considered as the specific case of imperfect repair, has a clear "physical" meaning and, consequently, the Brown and Proschan[3] imperfect repair model (see the next section) as well. This is because a "tiny" maintenance of the failed large system obviously keeps its state, at least, approximately the same as it was prior the failure.

The second stream deals with the, so-called, age correspondence problem that arises when modeling and comparing the performance of an item that can operate in different environments. The natural question is how to relate its age in one environment to that in the other (e.g, severer)? This is usually performed via the cumulative exposure model that establishes the required age correspondence, which gives rise to another, well established in the literature notion of virtual age. In accordance with this model, the equivalent ages for two regimes are obtained by equating the probabilities of failures in these regimes. This equivalence is widely used in accelerated life testing literature and practice (see, e.g., Nelson,[4,5] Meeker and Escobar,[6] to name a few).

Another question closely related to the described age correspondence problem is: how to model the performance of an item with one regime switched to the other at some point? We argue that, similar to the described case of virtual age for repairable items, the corresponding assumptions that allow to recalculate the corresponding age for this case are not sufficiently justified and analyzed in the literature and are just taken for granted. However, the proper understanding of relevant limitations of the models is essential. This will also be considered in what follows.

As far as we know, no results on virtual age modeling for items from heterogeneous populations were reported in the literature so far. However, one can hardly find homogeneous populations in real life and this assumption is usually made for simplicity and mathematical tractability. The corresponding discussion in our article, is probably the first step in this direction.

The article is organized as follows. In Section 2, the detailed discussion on virtual age in imperfect repair modeling is presented. Section 3 is devoted to age correspondence in a general setup. In Section 4, we develop a novel approach to describing virtual age for systems subject to shocks and also propose a new notion of a random virtual age. In Sections 5 and 6, we discuss the notion of virtual age when switching of regimes takes place, whereas Section 7 is devoted to virtual age in heterogeneous populations of items. Finally, some concluding remarks are given in Section 8.

2 | VIRTUAL AGE AND IMPERFECT REPAIR

Consider first, a nonrepairable item that has been incepted into operation at $t = 0$. Let its lifetime T be described by the absolutely continuous cdf $F(t)$, the pdf $f(t)$, and the failure rate $\lambda(t)$. Let $\lambda(t)$ be strictly increasing, which also usually describes some *deterioration processes*. In this case, the value of the failure rate at t uniquely defines chronological age of the item, that is, $\lambda^{-1}(\lambda(t)) = t$, where $\lambda^{-1}(\cdot)$ is the corresponding inverse operation. Let this item be repairable now and fails at time t^*. Thus, we want to perform the corresponding corrective maintenance action, that is, the repair. It is well known that the perfect repair brings its age to 0, whereas the minimal repair[7] does not change the age and the distribution of remaining lifetime in this case is as if nothing had happened (no failure, no repair). Other option is when repair reduces the age of the item that was just prior to failure to some intermediate level τ, that is, $0 < \tau < t^*$ (age reduction). This is a specific case of *imperfect repair* as a rather *strong assumption* is made meaning that the “shape” of the failure rate remains the same and the corresponding point just “slides along the failure rate curve”.[8] The corresponding remaining lifetime after the described type of a repair is obviously defined by the following survival function

$$\overline{F}(t|\tau) = \frac{\overline{F}(t+\tau)}{\overline{F}(\tau)}, \tag{1}$$

where $\overline{F} \equiv 1 - F$ and t is, obviously, an argument of this function describing the time to the next failure. The well-known Kijima’s models of imperfect repair[1] for repairable items (recurrent events), were based on this assumption (see also,[2] where the similar notions were discussed independently and later re-established and developed for the case of the corresponding asymptotic properties in Finkelstein[9] and Liu et al.[10] Thus, in this way, the term "virtual age" naturally emerged as the reduced time τ after the repair, can be considered as such, although Malik[11] was probably the first to consider the corresponding age reduction operation. A comprehensive study of the model and its generalizations were performed in the influential paper by Doyen and Gaudoin.[12] A recent work by Mercier and Castro[13] presents rigorous stochastic comparisons between the age reduction models with degradation reduction models (see below). A survey on virtual age models based on (1) and its generalizations (Kijima-type modeling) can be found in Tanwar et al.[14] See also Dijoux et al[15] and Nguyen et al[16] and references therein, where a comprehensive analysis of the literature on the subject is performed. Lindqvist[17] have generalized the univariate virtual age/imperfect repair concept for repairable multicomponent systems.

We had specified (1) for imperfect corrective repair/maintenance (i.e., after the failure). Obviously, the similar approach can be applied to imperfect preventive repair (preventive maintenance) and most of our reasoning in this paper is also relevant for the imperfect preventive maintenance (PM).[18]

Imperfect repair that is defined by (1) was suggested as a "natural generalization" of the minimal repair concept. Formally, the corresponding survival function after the minimal repair is defined by (1), where $\tau = t^*$. It is well known[19] that this type of repair occurs *often* in practice when, for example, a small part of a large failed system is repaired/replaced, which is, of course, an approximate, but very common. A relevant example is a series system of n (large) components described by the IFR distributions. Another possibility is to replace the failed system by the statistically identical one that was operating for the same time but did not fail ("hot" standby). Another popular model of imperfect repair (although not following (1)) is the Brown-Proschan (1983) model when each failure is perfectly repaired with a given probability and is minimally repaired with a complementary probability. As both options have a clear practical meaning, so does the model itself.

However, general reduction of age to some intermediate level modeled by (1) does not have this practical, justified meaning and can be considered as plausible mathematical model, which possesses tractable properties, as, for example, in Finkelstein,[9] where some meaningful asymptotic properties were obtained for recurrent events governed by Kijima's model.

In order for the virtual age concept that arises in the problem of imperfect repair/maintenance to be sound and practically justified, there should be a clear description of the corresponding repair operations that result in (and conform with) relationship (1). In our opinion, this is not the case for most repairable systems except for a specific case of minimal repair.

On the other hand, although not practically justified in the sense defined above, this black-box reasoning can be effectively used, in principle, for fitting the real imperfect repair data, which is done in numerous publication (see Levitin and Lisniansky,[20] Dijoux et al,[15] de Toledo et al,[21] Liu et al,[10] to name a few). In fact, this is the case for many statistical applications when the model itself does not necessarily follow the real, for example, degradation processes which it models.

Observe that when, for example, the multicomponent system fails, its imperfect repair is usually performed as repair/replacement of the last failed component and possibly of some other components if they have failed before and did not cause a failure of a system. Then why would the corresponding remaining lifetime's failure rate in this case have the same shape (shifted) as that just before the failure, as defined by (1)? In principle, there are no physical or engineering justifications for this property (as we, obviously, have for the case of perfect or minimal repairs). Another example supporting our claim is in the area of software reliability, where, in accordance to numerous existing models, each debugging (that can be considered as an imperfect repair) changes the shape of the corresponding failure rate (see chapter 9 in Finkelstein[8] and references therein).

Remark 1. We can think probably about one possibility that can have some relevance to reality. Assume that our repair/maintenance is performed by replacement with the used item that had been operating for the time τ previously. Formally, this will be the repair that conforms with (1); however, a lot of assumptions and practical conditions should be in place for the corresponding justification. For example, the previous usage should be in the same environment, there should be a sufficient stock of items with different usage, etc.

Remark 2. Another popular model used for describing imperfect repair is the failure rate reduction model, when during the repair action the current and "remaining" failure rate is decreased in accordance with some rule (usually, proportional). There are also numerous papers on that mostly focusing on relevant inferential aspects (see Doyen and Gaudoin[12] and its follow-ups). However, distinct from the age reduction models, this type of repair is much better justified, as it does not require the same type of distribution after it and is based on a natural assumption that repair should decrease the failure rate. Needless to say that this approach goes back to the famous Cox PH model,[22] which is widely used in statistical analysis.

To illustrate our point, further, consider the "cold" standby system with n exponentially distributed i.i.d. lifetimes of components with failure rates λ, the corresponding failure rate of a system is an increasing function with $\lambda(0) = 0$ and $\lim_{t\to\infty}\lambda(t) = \lambda$, whereas the system failure rate[19] is given by

$$\lambda(t) = \frac{\lambda^n t^{n-1}}{(n-1)!\sum_{k=0}^{n-1}\frac{(\lambda t)^k}{k!}}. \tag{2}$$

Thus in practice, imperfect repair for this failed system (all n components had failed) means that only $1 < k < n$ components are replaced ($k = 1$ is the "minimal repair" and $k = n$ is the perfect repair). Obviously, after this type of repair, the shape of the failure rate changes and one cannot discuss this setting in terms of model (1).

Motivated by this example, we will briefly discuss now some aspects of imperfect repair modeling for systems with observed degradation. In fact, the number of operating units in the previous example can be also considered as the discrete variable of this degradation. Thus, assume that an item's degradation process is described by an *increasing* stochastic process $\{W_t, t \geq 0\}$ with *independent increments*. A failure occurs when this process reaches a predetermined (deterministic) level r. Then the cdf of the time to failure in this case is just $F_r(t) = \Pr(W_t \geq r)$. Imperfect repair means that not all deterioration is eliminated by the repair action. In line with the reasoning above, assume that the imperfect repair action results in reducing degradation to the level $0 \leq r_0 \leq r$. The perfect repair action in this case corresponds to $r_0 = 0$. Let, for definiteness, assume that $\{W_t, t \geq 0\}$, $W_0 = 0$ is a homogeneous gamma process with the shape function αt and the scale parameter λ. Then the time to failure distribution from the reduced degradation level $0 < r_0 < r$ is given by[23,24]

$$\begin{aligned} F_{r-r_0}(t) &= \Pr(W_t \geq r - r_0) \\ &= \int_{r-r_0}^{\infty} Ga(x|\alpha t), \lambda)dx = \frac{\Gamma(\alpha t, (r-r_0)\lambda)}{\Gamma(\alpha t)}, \end{aligned} \tag{3}$$

where $\Gamma(b, x)$ is an incomplete gamma function for $x \geq 0$, $b > 0$ defined as

$$\Gamma(b, x) = \int_{x}^{\infty} t^{b-1}\exp\{-t\}dt.$$

Thus, depending on the level r_0, after the imperfect repair of this kind, we have different distributions and different shapes of the corresponding failure rates for the (remaining) lifetime although defined via the gamma distributions (but with different parameters). Obviously, this is in contrast to (1), where the shape of the failure rate is the same and only the initial point τ defines the corresponding shift. Therefore, we cannot define the virtual age after the imperfect repair in the described sense. However, this model of imperfect repair has a clear practical meaning, that is, reduction of the observed degradation. On the other hand, this specific example can result in defining the virtual age of a different type that also can be justified. This will be done after discussing the notion of age correspondence in Sections 3 and 4. Also note that, relationship (3) in this simple form is valid only for homogeneous processes. Indeed, for the nonhomogeneous processes, for the proper definition of remaining lifetime after reduction of degradation, the corresponding time should be defined. This will be addressed in Section 4 while introducing the new notion of virtual age for degrading items.

An important difference in description of the above models is in the level of available/used information. In (1) the *black-box* description of a system is employed, whereas in the example of the one out of n system, we have additional information on the structure of the system, thus the corresponding imperfect repair can be referred to as *information-based imperfect* repair.[25,26] Similarly, the amount of degradation presents information on the state of a system. It should be noted that reliability structures of most engineering systems are known, whereas the black-box description is more relevant to items/components that form a system.

Apart from imperfect repair, there can be other settings where the concept of virtual age based on the corresponding shift in the distribution function/failure rate can arise. Moreover, as we will see, it can be justified in some cases.

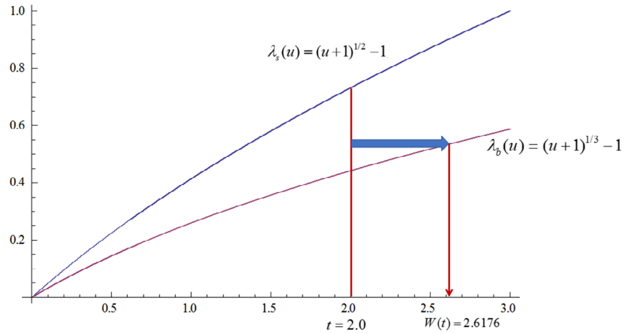

FIGURE 1 Example of age correspondence for $\lambda_s(u) = (u+1)^{1/2} - 1$, $\lambda_b(u) = (u+1)^{1/3} - 1$, and $t = 2.0$ [Color figure can be viewed at wileyonlinelibrary.com]

3 | AGE CORRESPONDENCE

Consider an item that operates in a baseline environment (regime) and denote its cdf of time to failure by $F_b(t)$. Let another statistically identical item operate, for example, in a severer environment with the cdf of time to failure denoted by $F_s(t)$. Denote by $\lambda_b(t)$ and $\lambda_s(t)$ the failure rates in two environments, respectively. We want to establish an age correspondence between the systems in two regimes by considering the baseline as a reference one. Assume that the lifetimes in two environments are ordered in the sense of the usual stochastic order[27] as

$$\overline{F}_s(t) \leq \overline{F}_b(t), \quad t \in (0, \infty). \tag{4}$$

Inequality (4) implies the following equation

$$F_s(t) = F_b(W(t)), \quad W(0) = 0, \ \ t \in (0, \infty), \tag{5}$$

which can be interpreted as a general accelerated life model (ALM) (see also Cox and Oakes,[28] Meeker and Escobar,[6] Finkelstein,[8] to name a few) with a time-dependent scale-transformation function $W(t) \geq t$. Note that $W(t)$ is an increasing function of time.

Relationship (5) can be viewed as an equation for obtaining the function $W(t)$, as

$$\exp\left\{-\int_0^t \lambda_s(u)du\right\} = \exp\left\{-\int_0^{W(t)} \lambda_b(u)du\right\}$$
$$\Rightarrow \int_0^t \lambda_s(u)du = \int_0^{W(t)} \lambda_b(u)du \tag{6}$$

and (6) can be already *interpreted* in terms of the cumulative exposure model given in Nelson.[5] Thus, age t in the severer regime is equivalent to age $W(t)$ in a milder one (the same fraction of units failing), which is illustrated by Figure 1.

Example 1. Let the failure rates in both regimes be increasing, positive power functions (the Weibull distributions), which are often used for lifetime modeling of degrading objects, that is,

$$\lambda_b(t) = \alpha t^{\beta}, \quad \lambda_s(t) = \mu t^{\eta} \quad \alpha, \beta, \mu, \eta > 0.$$

The function $W(t)$ is defined from (6) in the following way.[26]

$$W(t) = \left(\frac{\mu(\beta+1)}{\alpha(\eta+1)}\right)^{\frac{1}{\beta+1}} t^{\frac{\eta+1}{\beta+1}}.$$

As $W(t)$ is increasing, $W^{-1}(t)$ exists and is also increasing. Then, similar to (6),

$$\int_0^{W^{-1}(t)} \lambda_s(u)du = \int_0^t \lambda_b(u)du, \tag{7}$$

meaning that age t in the baseline regime is equivalent to age $W^{-1}(t)$ in a more severe one.

Then, for the considered example,

$$W^{-1}(t) = \left(\frac{\mu(\beta+1)}{\alpha(\eta+1)}\right)^{-\frac{1}{\eta+1}} t^{\frac{\beta+1}{\eta+1}}.$$

Note that, both functions $W(t)$, $W^{-1}(t)$ can be also interpreted as the corresponding virtual ages depending on which regime is considered to be the baseline one, that is,

- *the function $W(t) \geq t$ can be interpreted as the virtual age of an item in the milder regime that corresponds to the real age of an item that was operating for time t in the severer regime,*

whereas.

- *the function $W^{-1}(t) \leq t$ can be interpreted as the virtual age of an item in the severer regime that corresponds to the real age of an item that was operating for time t in the milder regime.*

While establishing the age correspondence for two regimes, we are assuming that these regimes are static. Note that, there can be other, more complex ways of establishing the relevant age correspondence for the case of random environments (see Reference [29], where the notion of "intrinsic age" that is similar to our age in the baseline environment was defined).

Example 2. Our main concern with respect to model (1) was that usually, in practice, there is no "engineering or physical evidence" for the remaining lifetime to behave as given by this model, although it can be a plausible approximation from the inferential point of view. On the other hand, it might be more realistic at certain instances that the imperfect repair keeps *the type of the lifetime distribution unchanged*, whereas one or more parameters can be changed. As an example, consider the gamma process case defined by (3) and assume that this process is homogeneous, that is, $\alpha(t) = \alpha\, t$. We see that after the imperfect repair, only parameter $r\lambda$ had changed to $(r - r_0)\lambda$. Thus, in accordance with the age correspondence principle, the virtual age for an item with the failure threshold $(r - r_0)$ can be found similar to (7) as the solution to the following equation:

$$F_{r-r_0}(W^{-1}(t)) = F_r(t), \quad r_0 < r, \tag{8}$$

where, obviously, $W^{-1}(t) \leq t$. We formally do not have two regimes here, although the case with the smaller threshold can be *interpreted* as the severer regime. This also follows from our general reasoning in (4)-(6), as the corresponding lifetimes (the initial lifetime and the remaining lifetime after the described imperfect repair) are ordered in the sense of the usual stochastic order[27] as

$$F_{r-r_0}(t) \geq F_r(t),$$

which goes in line with Finkelstein[30] and is also due to the assumption that the process is homogeneous. We will deal with the nonhomogeneous case at the end of the next section.

Remark 3. In Finkelstein,[8,26] the function $W(t)$ was called "the statistical virtual age", meaning that it was obtained from Equations (5) to (6), establishing the corresponding "statistical equivalence," whereas the virtual age for regime switching was called "recalculated virtual age". This had caused some terminological controversy in further publications. Therefore, in this article, we follow a more direct terminology.

4 | SHOCKS AND AGE CORRESPONDENCE

To the best of our knowledge, virtual age concept in the context of shocks modeling was not really widely implemented in reliability literature. One can probably mention the recent paper by Qiu and Cui,[31] where a system affected by shocks was considered. These authors assumed that each shock increases the virtual age of a system on a random amount. Virtual age was defined via the baseline cdf/failure rate and the corresponding shifts. However, similar to the case of imperfect repair modeled by (1), no justification of this assumption was provided. In Arnold et al,[32] the effect of shocks on one-component and multicomponent repairable systems was modeled. The suggested approach has two sources that result in imperfect repair. First, the failures of a component are imperfectly repaired in accordance with Kijima models. Second, each shock acts as a multiplier for the corresponding remaining lifetime decreasing the latter. We have a guess that this multiplier (under certain assumptions) can be interpreted/justified as the decrease in the corresponding virtual age, similar to how it is performed in this section. However, this needs further study.

In what follows, we will describe a model with virtual age justified via the age correspondence principle for a slightly different setting when each shock can result in the immediate failure of a system. It should be noted that as shocks constitute the severer environment, the age correspondence principle is relevant.

Let $\{N(t), t \geq 0\}$ be the NHPP with rate $r(t)$ and consider the corresponding extreme shock model, that is, each shock results in a failure of an item with probability $p(t)$ (and is survived without any consequences with probability $q(t) = 1 - p(t)$). Then, it is well known[33], that if the time to failure (without shocks) and the shock process are independent, then the corresponding survival function for an item subject to shocks is defined as

$$\overline{F}_s(t) = \overline{F}_b(t) \exp\left\{ -\int_0^t p(u)r(u)du \right\} = \overline{F}_b(W(t)). \tag{9}$$

Indeed, the baseline (milder) environment now is the one without shocks, whereas the severer environment is described additionally by a shock process affecting an item. Then, similar to the previous section, the corresponding virtual age $W(t)$ can be obtained from this equation. Thus, for example, if an item had survived under a shock process in [0, t), its corresponding virtual age in the environment without shocks is $W(t)$.

How should we understand this statement made under the formal assumption of the model (4)-(5)? It should be done just in the population-wise sense in terms of proportions of items failed that are equal, that is, the decrease in the lifetime of an item due to shocks can be considered as the corresponding deterioration (population-wise). We will also use the degradation-based approach when considering the switching of regimes in Section 6.

The function $W(t)$ obtained from (9) defines the virtual age of an item subject to the NHPP of shocks. But what happens under *a single* shock? This can be considered in a more detailed way. Let an item with a continuous lifetime, described by the cdf $F_b(t)$, be affected by a single "instantaneous" shock at time $t = t^*$. Let probability that this will result in failure be $p(p(t^*)$ for a more general case) and probability that it will survive with no consequences be $q = 1 - p$, which describes a standard extreme shock model[33]. Assume now that the effect of shock can be equivalently interpreted as an immediate jump in the age of an item on some $\tau(t^*)$ that can be uniquely defined/justified via p as

$$F_b(t^* + \tau(t^*)) - F_b(t^*) = p(1 - F_b(t^*)). \tag{10}$$

Thus $t^* + \tau(t^*)$ can be also interpreted as the virtual age of an item (compared with the chronological age t^*) after a shock. The corresponding piecewise continuous cdf $F_s(t)$ in this case is (Figure 2)

$$F_s(t) = \begin{cases} F_b(t), \quad t < t^* \\ F_b(t^*) + (1 - F_b(t^*))p, \quad t = t^* \\ F_b(t + \tau(t^*)), t > t^* \end{cases}, \tag{11}$$

where the "horizontal-parallel" shift to the left on $\tau(t^*)$ of the original $F_b(t)$ was performed for $t > t^* + \tau(t^*)$. It can be easily seen that the remaining lifetimes after the vertical jump at t^* defined by the cdf (11) and after the corresponding horizontal jump in age (resulting in the virtual age $t^* + \tau(t^*)$) are described by the same cdf. Obviously, in this case, (5) holds with

$$W(t) = \begin{cases} t, \quad t < t^* \\ t + \tau(t^*), t \geq t^* \end{cases} \tag{12}$$

and the corresponding virtual age model is justified via the age correspondence principle.

Thus, the virtual age is *uniquely defined* by p and the baseline distribution in this setting. Note that parameter p can be effectively obtained from tests, field data, experiments, etc. and thus the virtual age, *in contrast to the imperfect repair model* (1), can be justified. The described approach in a similar manner can be generalized to the case of the fixed number of shocks affecting an item, after properly adjusting the $F_s(t)$ at each shock. On the other hand, for the NHPP process of shocks, the alternative way of obtaining virtual age via the age correspondence model is given by (9). This virtual age can be also justified, as both $F_s(t)$ and $F_b(t)$ can be estimated in practice.

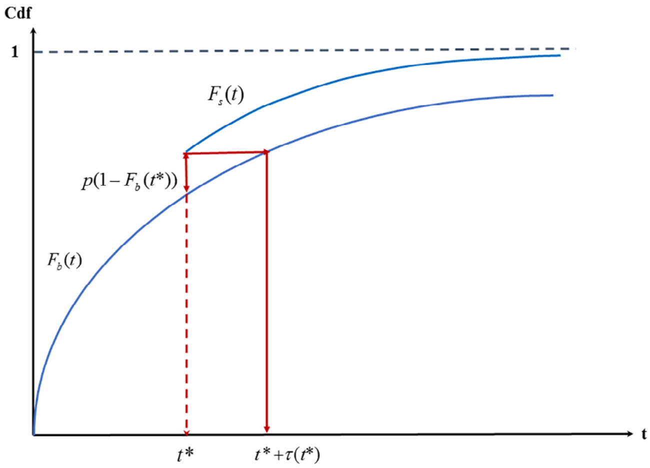

FIGURE 2 The graph for $F_s(t)$ [Color figure can be viewed at wileyonlinelibrary.com]

In sections to follow, we will show that the age correspondence principle can be used for individual items, when the regime of operation is switched to another one at some time point. It is important to note that obviously, the switching is performed for an operating component and there is no need to take into account probability of survival as in (10)-(11).

Consider now a situation when a shock results in a change of the lifetime distribution after its occurrence (see also Example 2 and the corresponding description in the Introduction). Assume that a shock effecting an operating item at t^* is survived and results in additional deterioration r_s. Let an item's deterioration just before the shock be r_d, $r_s + r_d < r$. Thus, in accordance with (3), the remaining lifetime without a shock is $F_{r-r_d}(t)$, whereas after the shock, it is defined by $F_{r-r_d-r_s}(t)$. Here, defining the corresponding remaining lifetimes, we assume as in (3) for simplicity that the underlying stochastic process of deterioration is homogeneous. Therefore, similar to Example 2, we can define the corresponding virtual age (just after a shock, as before it they are the same for the baseline regime and the one with a shock)) as

$$F_{r-r_d-r_s}(t) = F_{r-r_d}(W(t))$$

meaning that the virtual age of an operating item in t units of time after a shock at t^* is $t^* + W(t)$, whereas the calendar age for a unit without a shock is, obviously, $t^* + t$, $W(t) \geq t$.

After the foregoing discussion, we can come back to the described in Example 2 the degradation-based imperfect repair. It was shown that the ordering of remaining lifetimes after reducing the initial failure threshold from r to r_0, that is, $F_{r-r_0}(t) \geq F_r(t)$ for the corresponding distributions, holds only for homogeneous processes of degradation. What happens when the process of degradation of an item, $\{W_t, t \geq 0\}$, $W_0 = 0$ (with independent increments) is nonhomogeneous? In this case, the corresponding remaining lifetime is defined by the survival function

$$\overline{F}_{r-r_0}(t) = \int_0^\infty P(W_{x+t} - W_x \leq r - r_0) f(x, r_0) dx,$$

where $f(x, r_0) = \frac{\partial}{\partial x} F(x, r_0)$ and $\overline{F}(t, r_0) = P(W_t \leq r_0)$ is a random time it takes for the process $\{W_t, t \geq 0\}$, $W_0 = 0$ to accumulate degradation r_0. We will call this random variable the *degradation-based virtual* age, which, as described, has a clear justified meaning. We plan to further investigate it in the future research. Obviously, it has a more clear, specific meaning for the given setting than deterministic $W(t)$ obtained via the general age correspondence principle.

5 | STATE-DEPENDENT VIRTUAL AGE FOR REGIMES SWITCHING

In the previous section, we have considered the age correspondence for statistically identical items operating in different regimes. This age correspondence is widely used in accelerated life testing, as it allows (under certain assumptions) to perform reliability tests in a much shorter testing time.

Our setting in this section is different and considers switching of a regime of an operating system to a different one and, accordingly, the virtual age now depends on the state after the switching. We want to answer a question: can we use the reasoning of the previous section in this case? More specifically, let a system that has been already operating in a baseline regime for the time τ, be switched to a severer regime at $t=0$ (it is just more convenient now to perform switching at $t = 0$). On the other hand, in accordance with (7), if an item operates during the time $W^{-1}(\tau)$ in the severer regime before $t = 0$ (age correspondence), then its future performance is statistically the same as if it were operating during τ in the baseline regime before the switch. For simplicity, we assume here that both regimes are time-independent (constant loads). The function $V_s(\tau) \equiv W^{-1}(\tau)$ in this case is also called the virtual age (as discussed above). Thus, in accordance with this reasoning and relationship (1) (see Cha et al[34] for more details), we can write the expression for the survival function of the remaining lifetime after the switching as

$$\overline{F}_s(t|V_s(\tau)) = \frac{\overline{F}_s(t + W^{-1}(\tau))}{\overline{F}_s(W^{-1}(\tau))} = \exp\left\{-\int_0^t \lambda_s(u + W^{-1}(\tau))du\right\}. \tag{13}$$

Relationship (13) describes the, so-called, *black-box* scenario when the system's structure is not known (or does not exist). We can also obviously use cumulative exposure model for obtaining the age correspondence for systems operating in different environment. However, when we switch the regime for systems that can operate in different, clearly defined states, using the cumulative exposure model does not guarantee that the system will be in the same state before and after the switching. Therefore, either virtual age in this case should be defined in a different way, or it is unreasonable to use this concept for this case.

As an illustration of this statement, we now consider a system of n components having different states defined by the numbers and combinations of operating components. Let it be, specifically, as in Section 2, the "cold" standby system with n exponentially distributed i.i.d. lifetimes of components with failure rates λ_b in the baseline regime and $\lambda_s(\lambda_s > \lambda_b)$, in the severer regime. Alternatively, more general Markovian models of systems can be also considered in the same line.

Remark 4. Note that the regime modeled by a linear scale transformation $W(t) \equiv wt$ does not imply the constant in time proportional hazards (PH) model (it acts time-dependent on the corresponding failure rates) as.

$$\lambda_s(t) = w\lambda_b(wt)$$

and can result in proportionality only for the Weibull distributions. On the other hand, which is obvious from this equation, if $\lambda_s(t) \equiv \lambda_s$ and $\lambda_b(t) \equiv \lambda_b$, then $w = \lambda_s/\lambda_b$.

Thus, the severer environment "acts" with the factor $w = \lambda_s/\lambda_b$ on the baseline failure rate of each operating component. Define by $P_m(t, \lambda_b)$ and $P_m(t, \lambda_s)$, $m = 0, 1, \dots, n$ probabilities that our system is in the state with $m = 0, 1, \dots, n-1$ failed components, for both regimes, respectively. Obviously,

$$P_m(t, \lambda_b) = \exp\{-\lambda_b t\}\frac{(\lambda_b t)^m}{m!}; \quad P_m(t, \lambda_s) = \exp\{-\lambda_s t\}\frac{(\lambda_s t)^m}{m!}. \tag{14}$$

Further performance of the system after switching depends only on the state (which is the same after switching). Let now equate probabilities *for each state*, similar to what was done in the black-box reasoning without considering states

$$\widetilde{P}_m(\tau, \lambda_b) = \widetilde{P}_m(V_m(\tau), \lambda_s), \quad m = 0, 1, \dots, n-1, \tag{15}$$

where

$$\widetilde{P}_m(\tau, \lambda) = \frac{P_m(\tau, \lambda)}{\sum_{k=0}^{n-1} P_k(\tau, \lambda)}$$

is the conditional probability that given that the system is operable at $t = 0$, it is in the state m. Relationship (15), which is based on the general age correspondence principle (5)-(6), means that conditional probabilities of being in state m for a system operating in $[-\tau, 0)$ in the baseline regime are the same as for this system operating in $[-V_m(\tau), 0)$ in the severer regime. Thus, we define $V_m(\tau)$ as the state-dependent virtual age in this case. For our setting, it follows from (14) to (15) that

$$V_0(\tau) = V_1(\tau) = \cdots = V_{n-1}(\tau) \equiv \widetilde{V}(\tau) = \frac{\lambda_b}{\lambda_s}\tau. \tag{16}$$

The virtual age $W^{-1}(\tau)$ for this specific system can be obtained from

$$\int_0^{W^{-1}(\tau)} \frac{{\lambda_s}^n u^{n-1}}{(n-1)!\sum_{k=0}^{n-1}\frac{(\lambda_s u)^k}{k!}}du = \int_0^{\tau} \frac{{\lambda_b}^n u^{n-1}}{(n-1)!\sum_{k=0}^{n-1}\frac{(\lambda_b u)^k}{k!}}du. \tag{17}$$

It can be easily seen by changing the variable of integration in the right-hand side of (17) to $y = (\lambda_b/\lambda_s)u$ that $W^{-1}(\tau) = \widetilde{V}(\tau)$, which could be expected for the described system with identical components and identical constant change of the load coefficient λ_s/λ_b. Thus, the virtual age defined via the general age-correspondence equations (6)-(7) and the state-dependent virtual age coincide. Obviously, this holds for *other systems with exponentially distributed i.i.d. components*. To see this, note that the corresponding cdfs can be viewed as functions of a variable λt. Denote the system cdf by $F(t)$ when $\lambda = 1$. Then,

$$F_s(t) \equiv F(\lambda_s t) = F(\lambda_b(\lambda_s/\lambda_b)t) \equiv F_b((\lambda_s/\lambda_b)t),$$

where $W(t) = (\lambda_s/\lambda_b)t$; $W^{-1}(t) = (\lambda_b/\lambda_s)t$ and everything is defined by the same linear transformation of time when the baseline regime is switched to the severer one.

Situation is different for the nonidentical independent components, as the coefficients $\lambda_{i,\,b}/\lambda_{i,\,s}$, $i = 1, 2, \ldots$, can vary now, as the impact of regime/environment can change from component to component. For illustration, consider a parallel system of two independent components.

Example 3. Let the failure rates of these components be $\lambda_{i,\,b}$, $\lambda_{i,\,s}$, $i = 1, 2$, respectively and $\lambda_{1,\,b}/\lambda_{1,\,s} \neq \lambda_{2,\,b}/\lambda_{2,\,s}$. We have 3 states now with the corresponding probabilities

$$\widetilde{P}_i(\tau, \lambda_b) = \frac{P_i(\tau, \lambda_b)}{\sum_{k=0}^{2} P_k(\tau, \lambda_b)}, \quad i = 0, 1, 2, \tag{18}$$

where

$$\begin{aligned}
P_0(\tau, \lambda_b) &= \exp\{-(\lambda_{1,b} + \lambda_{2,b})\tau\},\\
P_1(\tau, \lambda_b) &= \exp\{-\lambda_{1,b}\tau\}(1 - \exp\{-\lambda_{2,b}\tau\}),\\
P_2(\tau, \lambda_b) &= \exp\{-\lambda_{2,b}\tau\}(1 - \exp\{-\lambda_{1,b}\tau\}).
\end{aligned}$$

Similar to (15), the equations for obtaining state-dependent virtual ages $V_0(\tau)$, $V_1(\tau)$, and $V_2(\tau)$ are

$$\widetilde{P}_i(\tau, \lambda_b) = \widetilde{P}_i(V_i(\tau), \lambda_s), \quad i = 0, 1, 2. \tag{19}$$

Obviously, in this case

$$V_0(\tau) \neq V_1(\tau) \neq V_2(\tau) \neq \widetilde{V}(\tau).$$

Thus, each state has its own virtual age.

Consider now a simplest case of a system with arbitrary distributions for lifetimes of components, that is, a cold standby system with identical components described by the cdfs $F_b(t)$ and $F_s(t)$; $F_s(t) = F_b(W(t))$. The state of an operating at $t = 0$ system is described now by the number of failed components, $i = 0, 1, 2, \ldots, n-1$ and the time spent in the current state. Thus, it is not feasible to define one virtual age (for each state) even in this simplest system with arbitrary distributions. Thus, as it is shown using simple examples, the state-dependent virtual age can be a reasonable approach only for Markovian systems (with exponentially distributed lifetimes of components), where it provides more justified reasoning as compared with the black-box approach.

In the next section, we describe and *interpret* the black-box approach that is considered on the whole time axis.[8,26] Under certain assumptions, this virtual age will be formally introduced via the transformation of the corresponding distribution functions. The main difference between the black-box virtual age used in (13) and that in the following section is in conditioning. In (13), we condition on the event that the switching of regimes at $t = 0$ is performed for the operating item. In what follows, there is no conditioning of this kind and the switching is performed in the *future* if an item survives up to this time. This is the important methodological difference.

6 | PERFORMANCE ON THE WHOLE TIME AXIS

Consider now an item that starts operation at $t = 0$ in the baseline regime and is switched to the severer regime at $t = x$. Both regimes are defined in $[0, \infty)$. Assume that $W(t)$ in (10) is differentiable, thus

$$W(t) = \int_0^t w(u)du$$

and interpret $w(t)$ as the rate of degradation in the severer regime, whereas it is 1 in the baseline regime. Thus, $w(t) \geq t$, in fact, has a meaning of a relative degradation. Let the cdf of the corresponding lifetime can be written as.[26]

$$F_{bs}(t) = \begin{cases} F_b(t), & 0 \leq t < x, \\ F_b\left(x + \int_x^t w(u)du\right), & x \leq t < \infty. \end{cases} \tag{20}$$

Furthermore, we make a *formal* transformation of the second row on the right-hand side of this equation, that is,

$$\begin{aligned} F_b\left(x + \int_x^t w(u)du\right) &= F_b\left(\int_{\tau(x)}^t w(u)du\right) \\ &= F_b(W(t) - W(\tau(x))). \end{aligned} \tag{21}$$

From (21), $\tau(x) < x$ is uniquely defined from the equation

$$x = \int_{\tau(x)}^x w(u)du = W(x) - W(\tau(x)). \tag{22}$$

Thus, the probability of failure of an item operating in a baseline regime in $[0, x)$ is equal to the probability of failure of this item that would have been incepted into operation at $t = \tau(x) < x$ under the severer regime. Recall that we assume that both regimes are time-independent. Therefore, hypothetical prolongation of the severer regime backwards (from the switching point) is well justified. Thus, formally $\hat{V}(x) \equiv x - \tau(x)$ can be viewed as the corresponding virtual age *for this model* (on the whole axis).

It follows from (22) that

$$\hat{V}(x) = x - W^{-1}(W(x) - x). \tag{23}$$

It was shown in Finkelstein[8] that $\hat{V}(x) = W^{-1}(x)$ only for the case of the *linear function* $W(t) = wt$. The last statement is also clear from intuitive considerations as in this case, as the impact of the severer regime is time-independent ($w(t) \equiv w$), it does not matter where the interval $[\tau(x), x)$ is placed and only its duration matters. This reasoning obviously depends on the understanding of the severer regime before and after the switching. However, assuming constant in time regimes makes the reasoning simpler by assuming, as mentioned hypothetical prolongation of the regime backwards.

Example 4. (Finkelstein[8]). Let $W(t) = wt$, $w > 1$. Then

$$x = \int_{\tau(x)}^x wdu \Rightarrow \tau(x) = \frac{x(w-1)}{w}$$

and

$$W^{-1}(x) = x - \tau(x) = x/w; \quad W(x) = xw.$$

Thus, the notion of virtual age in this section has a supplementary meaning, as the future behavior of an item can be defined without it, using directly the cdf (21). In contrast to that, in the previous sections, we could not describe the future performance of an item (after switching) without defining a virtual age of some kind.

7 | VIRTUAL AGE AND HETEROGENEOUS POPULATIONS

In order to emphasize our discussion of model (1) in Section 2, consider now an important in practice case of heterogeneous populations of items. Indeed, due to various reasons (e.g., nonstability in production processes, environmental changes, human errors, etc.), in practice, engineering items are heterogeneous.[35] We will briefly discuss the virtual age concept that results from the age correspondence in this case as well.

For illustration, consider a mixture of subpopulations described by the continuous frailty variable Z with the pdf $\pi(z)$ and a subpopulation failure rate (given $Z = z$) defined by the popular proportional model $z\lambda_b(t)$, where $\lambda_b(t)$ is some baseline failure rate. Then, the population survival function is given by

$$\overline{F}_m(t) = \int_0^\infty \exp\left\{-z\int_0^t \lambda_b(u)du\right\}\pi(z)dz = \int_0^\infty \overline{F}_z(t)\pi(z)dz, \tag{24}$$

where $\overline{F}_z(t) = \exp\left\{-z\int_0^t \lambda_b(u)du\right\}$. Let an item randomly selected from the mixed population start at time 0 and, at time t^*, a repair/maintenance action is performed in accordance with model (1) reducing its age to τ, $\tau < t^*$. Then, the remaining lifetime of an item after this operation can be defined as

$$\frac{\overline{F}_m(\tau + t)}{\overline{F}_m(\tau)} = \frac{\int_0^\infty \overline{F}_z(\tau + t)\pi(z)dz}{\int_0^\infty \overline{F}_z(\tau)\pi(z)dz} = \int_0^\infty \frac{\overline{F}_z(\tau + t)}{\overline{F}_z(\tau)} \frac{\overline{F}_z(\tau)\pi(z)}{\int_0^\infty \overline{F}_w(\tau)\pi(w)dw}dz. \tag{25}$$

Thus, the virtual age τ is defined via the population distribution $\overline{F}_m(\tau)$. However, the maintenance action is performed on an actual item, not on a hypothetical one picked up from a population. Thus, survival function of the remaining lifetime, given $Z = z$ is $\overline{F}_z(\tau + t)/\overline{F}_z(\tau)$, whereas the corresponding conditional distribution of frailty $Z \mid T > t^*$ is

$$\frac{\overline{F}_z(t^*)\pi(z)}{\int_0^\infty \overline{F}_w(t^*)\pi(w)dw}.$$

Therefore, in this case, the unconditional remaining lifetime of an item is defined by the following survival function.

$$\int_0^\infty \frac{\overline{F}_z(\tau + t)}{\overline{F}_z(\tau)} \frac{\overline{F}_z(t^*)\pi(z)}{\int_0^\infty \overline{F}_w(t^*)\pi(w)dw}dz, \tag{26}$$

which is, obviously, different from (25). Thus, additionally to our concerns in Section 1, we have a certain *ambiguity* in defining the virtual age in heterogeneous populations.

Remark 5. It is easy to compare the remaining lifetimes described by (25) and (26), accordingly. Indeed, the likelihood ratio of the corresponding mixing random variables $Z \mid T > \tau$ and $Z \mid T > t^*$,

$$\exp\left\{z\int_\tau^{t^*} \lambda_b(u)du\right\} \frac{\int_0^\infty \overline{F}_w(t^*)\pi(w)dw}{\int_0^\infty \overline{F}_w(\tau)\pi(w)dw}$$

is increasing in z, which means that $Z \mid T > \tau$ is larger than $Z \mid T > t^*$ in this sense, and therefore, in the sense of the usual stochastic order[27]. Therefore, similar to Finkelstein,[8]

$$\int_0^\infty \frac{\overline{F}_z(\tau+t)}{\overline{F}_z(\tau)} \frac{\overline{F}_z(\tau)\pi(z)}{\int_0^\infty \overline{F}_w(\tau)\pi(w)dw} dz < \int_0^\infty \frac{\overline{F}_z(\tau+t)}{\overline{F}_z(\tau)} \frac{\overline{F}_z(t^*)\pi(z)}{\int_0^\infty \overline{F}_w(t^*)\pi(w)dw} dz.$$

Thus, the population virtual age model (25) underestimates virtual age defined on the subpopulation level in (26).

On the other hand, age correspondence for heterogeneous populations is defined *nonambiguously* via the population characteristics. Assume that components from a heterogeneous population described by the pdf $\pi(z)$ can operate in two regimes. Then, similar to (24), we have for the population distributions

$$\overline{F}_{mb}(t) = \int_0^\infty \exp\left\{-z\int_0^t \lambda_b(u)du\right\}\pi(z)dz \equiv \int_0^\infty \overline{F}_{zb}(t)\pi(z)dz,$$

and

$$\overline{F}_{ms}(t) = \int_0^\infty \exp\left\{-z\int_0^t \lambda_s(u)du\right\}\pi(z)dz \equiv \int_0^\infty \overline{F}_{zs}(t)\pi(z)dz,$$

where $\lambda_b(t) < \lambda_s(t), \forall\, t > 0$. As $\overline{F}_{mb}(t) \geq \overline{F}_{ms}(t)$, we can use (4)-(5) for obtaining the virtual age $W(t)$ as it was done for the homogeneous case.

8 | CONCLUDING REMARKS

We discuss in different settings important notions of virtual age that are widely used in the literature, however, often without questioning the underlined assumptions and the meaning of these assumptions as well. Our main interest is in the fundamental underlying properties of this notion in different setups.

In order for the virtual age concept that arises in the problem of imperfect repair to be sound and practically justified, there should be a clear description of the corresponding repair operations that results in (and conforms with) relationship (1). In our opinion, this is not the case for most repairable systems except a specific case of minimal repair.

On the other hand, the new virtual age model for systems subject to external shocks is suggested and justified via the probabilities of failures on shocks. It also implicitly uses the age correspondence for statistically identical components operating in different environments. Moreover, a random virtual age for a degrading system to accumulate the predetermined level of degradation is proposed. This notion should be studied in the future research.

It is shown using simple examples, that the state-dependent virtual age can be a reasonable approach only for Markovian systems (with exponentially distributed lifetimes of components), where it provides more justified reasoning compared with the black-box approach.

Under certain assumptions, the black-box virtual age is formally introduced via the transformation of the corresponding distribution functions. However, in this context, it has only a supplementary meaning as future performance (after switching) of an item can be described without it.

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DATA AVAILABILITY STATEMENT

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ORCID

Maxim Finkelstein https://orcid.org/0000-0002-3018-8353
Ji Hwan Cha https://orcid.org/0000-0002-9158-6275

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